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# Study of various Network commands used in Linux and Windows

**Aim:-** To study various Network commands used in Linux and Windows.

## Basic Networking Commands:-

**arp -a:-** address resolution protocol. shows IP and MAC address of router.

**hostname:-** TCP/IP command that displays name of computer

**ipconfig/all:-** Gives detailed configuration information about TCP/IP connection.

**nbtstat -a:-** solves problems with NetBIOS name resolution.

**net stat:-** displays statistics of active TCP/IP connections, interface statistics, etc.

**nslookup:-** (name server lookup) performs DNS lookups in linux to display DNS details. It operates in 2 modes- interactive, non-interactive.

pathping:- unique to windows, combination of ping and tracert commands. Launches a 25 second test of each router.

ping:- test connectivity between nodes. Uses ICMP to communicate to other devices.

Route:- Used to show/manipulate the IP routing table. To setup static routes via an interface.

## Important Linux networking commands:-

1. ip

Basic commands for setting up new systems, troubleshooting and seeing address info, network devices, tunnels, etc.

ip <options> <object> <command>

2. ifconfig

main command for sys admin for configuring and troubleshooting networks. later replaced by ip.

3. mtr

(Matt's traceroute), a program in CLI for network diagnostic and troubleshooting. combines ping and traceroute. provides statistics of each hop, response time, etc.

mtr <options> hostname/IP

## 4. tcpdump

For displaying and capturing packets. needs root access before capture.

installation command:- dnf install -y tcpdump.

before capture :- tcpdump - D

for traffic on eth0 :- tcpdump -i eth0.

capture traffic to and from 1 host:-

tcpdump -i eth0 -c 10 host 8.8.8.8

for traffic from 8.8.8.8 :-

tcpdump -i eth0 src host 8.8.8.8

To and from network:-

tcpdump -i eth0 net 10.1.0.0 mask 255.255.255.0

(or)

tcpdump -i eth0 net 10.1.0.0/24

To and from port numbers:-

tcpdump -i eth0 port 53

h0 = 8.8.8.8 and port 53

and port 443

5. ping :-

tool to verify IP-level connectivity to another TCP/IP by sending ICMP Echo request messages. receipt of echo is displayed. Used to trouble-shoot connectivity, reachability and name resolution.

ex:- ping google.com

=> ping google.com (216.58.206.174) 56 (84) bytes of data.

64 bytes from sof02s27-in-f14.1e100.net (216.58.206.174):

icmp_seq=1 ttl=56 time=10.7 ms;

.....

Stopped by CTRL + C.

if needed to ping 10 times :-

ping -c 10 google.com

## Configuring Ethernet connection by using nmcli

can manage connection's settings in CLI using nmcli utility.

Procedure :-

1. List network manager connection profile

=> nmcli connection show

2. nmcli connection add con-name <connection-name> ifname <device-name> type ethernet

3. Display current settings

nmcli connection show

4. Configure IPv4 settings:-

edit DHCP by:-

nmcli connection modify "Wired connection 1"
ipv4.method auto

(Skip if method → auto)

set static ipv4 address, etc:-

nmcli connection modify "Wired connection 1"
ipv4.method manual ipv4.addresses 192.0.2.1/24
ipv4.gateway 192.0.2.254 ipv4.dns 192.0.2.200
ipv4.dns-search example.com

5. Configure IPv6 settings:-

DHCP:-
command similar to IPv4

set static IPv6 address, etc:-

nmcli connection modify "Wired connection 1" ipv6.method
manual ipv6.addresses 2001:db8:1::fffe/64
ipv6.gateway 2001:db8:1::fffe ipv6.dns
2001:db8:1::ffbb ipv6.dns-search example.com

6. Activate profile:-

nmcli connection up Internal-LAN

## Verification :-

1. Display IP setting of NIC:-

ip address show enp1s0

2. Display IPv4 default :-

ip route show default

3. Display IPv6 default :-

ip -6 route show default

4. Display DNS :-

cat /etc/resolv.conf

5. Use ping to verify that host can send packets to other hosts

ping <host_name_or_IP_address>

**Result** :- Hence, various networking commands in Linux and windows were studied and executed successfully.

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